

## Chapter 3 Answer Key

Introduction to Statistics — MATH& 146

### Section 3.1 Exercises

**3.1.1.** For a *continuous* random variable, probability equals area under the density curve. A single point such as  $X = 2.5$  has no width, so the area above it is zero; hence  $P(X = 2.5) = 0$ . For a *discrete* random variable, each individual value is assigned a specific probability mass directly — no area is involved — so  $P(X = 2.5)$  is a perfectly meaningful number (though it may equal zero if 2.5 is not in the support).

**3.1.2.** The description specifies that the curve rises linearly from  $x = 0$  and reaches its peak at  $x = 2$ , so the peak is at  $\mathbf{x = 2}$ . To find the height: the triangular region has base = 4 (from  $x = 0$  to  $x = 4$ ). For total area = 1:

$$\frac{1}{2} \times \text{base} \times \text{height} = \frac{1}{2}(4)(h) = 1 \implies h = \frac{1}{2}.$$

The peak height is  $\mathbf{1/2}$ .

**3.1.3. False.** For a continuous distribution,  $P(X = 3) = 0$ , so including or excluding the endpoint 3 makes no difference:

$$P(X > 3) = P(X \geq 3).$$

This contrasts with discrete distributions, where the distinction between  $>$  and  $\geq$  can matter.

### Section 3.2 Exercises

**3.2.1.**  $X \sim U(0, 1)$ , so  $f(x) = 1$  on  $[0, 1]$  and probability equals interval length.

(a)  $P(X < 0.3) = 0.3 - 0 = \mathbf{0.30}$

(b)  $P(0.2 < X < 0.8) = 0.8 - 0.2 = \mathbf{0.60}$

(c)  $P(X > 0.95) = 1 - 0.95 = \mathbf{0.05}$

**3.2.2.**  $X \sim U(5, 20)$ .

**Mean:**  $\mu = \frac{5 + 20}{2} = \mathbf{12.5}$

**Standard deviation:**  $\sigma = \frac{20 - 5}{\sqrt{12}} = \frac{15}{\sqrt{12}} = \frac{15}{2\sqrt{3}} = \frac{5\sqrt{3}}{2} \approx \mathbf{4.33}$

**Probability:**

$$P(8 \leq X \leq 15) = \frac{15 - 8}{20 - 5} = \frac{7}{15} \approx \mathbf{0.467}$$

The density curve is a flat rectangle of height  $\frac{1}{15}$  on  $[5, 20]$ ; the shaded region is the strip from  $x = 8$  to  $x = 15$ .

**3.2.3.**  $X \sim U(0, 30)$ .

$$P(X > 20) = \frac{30 - 20}{30 - 0} = \frac{10}{30} = \frac{1}{3} \approx \mathbf{0.333}.$$

**Expected delay:**  $\mu = \frac{0 + 30}{2} = \mathbf{15}$  minutes.

## Section 3.3 Exercises

**3.3.1.** Using the Empirical Rule (no table needed):

- (a)  $P(-1 \leq Z \leq 1) \approx \mathbf{0.68}$  (68% of area within  $\pm 1\sigma$ ).
- (b) By symmetry, 5% of the total area lies outside  $\pm 2\sigma$ , split equally between the two tails. Therefore  $P(Z > 2) \approx 0.05/2 = \mathbf{0.025}$ .
- (c) By symmetry, the interval  $[-3, 0]$  contains exactly half of the 99.7% that lies within  $\pm 3\sigma$ :  $P(-3 \leq Z \leq 0) \approx 0.997/2 \approx \mathbf{0.4985}$ .

**3.3.2.** The normal curve is *asymptotic*: its tails approach the horizontal axis but never reach it. They extend infinitely in both directions. Yet the area still equals exactly 1 because the tails become so thin so quickly that their total area is finite — the infinite series of vanishingly small contributions converges. It is the same reason the sum  $\frac{1}{2} + \frac{1}{4} + \frac{1}{8} + \cdots = 1$  even though there are infinitely many terms.

**3.3.3.** The standard normal curve has 95% of its area within  $\pm 2\sigma$  of the mean (Empirical Rule approximation). From the z-table, the exact value is  $P(-1.96 \leq Z \leq 1.96) = 0.9500$ , so the shaded region lies between  $z = -1.96$  and  $z = 1.96$ .

## Section 3.4 Exercises

For each part, *draw the curve, mark the z-value(s), shade the region, and estimate whether the answer should be above or below 0.5 before reading the table.*

**3.4.1.**

- (a)  $P(Z \leq 0.54)$ : row 0.5, column 0.04  $\Rightarrow \mathbf{0.7054}$ . (Above 0.5: 0.54 is right of center.  $\checkmark$ )
- (b)  $P(Z \leq -1.32)$ : by symmetry  $= 1 - P(Z \leq 1.32)$ . Row 1.3, col 0.02 gives 0.9066. Therefore:

$$P(Z \leq -1.32) = 1 - 0.9066 = \mathbf{0.0934}.$$

(Below 0.5:  $-1.32$  is left of center.  $\checkmark$ )

- (c)  $P(Z > 2.10) = 1 - P(Z \leq 2.10)$ . Row 2.1, col 0.00 gives 0.9821. Therefore:

$$P(Z > 2.10) = 1 - 0.9821 = \mathbf{0.0179}.$$

- (d)  $P(-0.75 \leq Z \leq 1.50) = P(Z \leq 1.50) - P(Z \leq -0.75)$ .

$P(Z \leq 1.50)$ : row 1.5, col 0.00 = 0.9332.

$P(Z \leq -0.75) = 1 - P(Z \leq 0.75)$ : row 0.7, col 0.05 gives 0.7734, so  $P(Z \leq -0.75) = 0.2266$ .

$$P(-0.75 \leq Z \leq 1.50) = 0.9332 - 0.2266 = \mathbf{0.7066}.$$

**3.4.2.** Finding  $z^*$  by searching the body of the table:

(a)  $P(Z \leq z^*) = 0.10$ . Searching for 0.1000: the closest entry is 0.1003 at  $z = -1.28$ .

$$z^* \approx -1.28.$$

(b)  $P(Z > z^*) = 0.05 \Rightarrow P(Z \leq z^*) = 0.95$ . Searching for 0.9500: the table gives 0.9495 at  $z = 1.64$  and 0.9505 at  $z = 1.65$ ; the midpoint is  $z = 1.645$ . In practice,

$$z^* \approx \mathbf{1.645} \quad (\text{or use } z^* = 1.65 \text{ if rounding to two decimal places}).$$

**3.4.3.** By symmetry of the standard normal:  $P(Z \leq -1.96) = 1 - P(Z \leq 1.96)$ .

From the table, row 1.9 col 0.06:  $P(Z \leq 1.96) = 0.9750$ .

Therefore  $P(Z \leq -1.96) = 1 - 0.9750 = 0.0250$ . ✓ (The two values are equal, confirming symmetry.)

$$P(-1.96 \leq Z \leq 1.96) = 0.9750 - 0.0250 = \mathbf{0.9500}.$$

The value  $z = 1.96$  is important because it is the critical value for a **95% confidence interval** and a two-tailed hypothesis test at the 5% significance level. It marks the boundary of the central 95% of the standard normal distribution; this is why it will appear in nearly every inference procedure in Chapter 5.

## Chapter 3 Review

**1.**  $X \sim U(7, 9)$  (coffee machine, ounces per cup).

(a)  $f(x) = \frac{1}{9-7} = \frac{1}{2} = \mathbf{0.5}$  for  $7 \leq x \leq 9$ , and  $f(x) = 0$  otherwise. The density curve is a flat rectangle of height 0.5 sitting on the interval  $[7, 9]$ .

(b)  $P(7.5 \leq X \leq 8.5) = \frac{8.5-7.5}{9-7} = \frac{1}{2} = \mathbf{0.50}$ . This is the area of a rectangle of width 1 and height  $1/2$ .

(c)  $\mu = \frac{7+9}{2} = \mathbf{8}$  oz.  $\sigma = \frac{9-7}{\sqrt{12}} = \frac{2}{\sqrt{12}} = \frac{1}{\sqrt{3}} \approx \mathbf{0.577}$  oz.

(d)  $P(X > 8.8) = \frac{9-8.8}{9-7} = \frac{0.2}{2} = \mathbf{0.10}$ .

**2.** Using the z-table:

(a)  $P(Z \leq 1.65)$ : row 1.6, col 0.05 = **0.9505**.

(b)  $P(Z > -0.42)$ : by symmetry,  $P(Z > -0.42) = P(Z \leq 0.42)$  (the normal curve is symmetric around 0, so the right tail of a negative value equals the left tail of the corresponding positive value). Row 0.4, col 0.02 = **0.6628**.

(c)  $P(1.00 \leq Z \leq 2.33) = P(Z \leq 2.33) - P(Z \leq 1.00)$ .

Row 2.3, col 0.03:  $P(Z \leq 2.33) = 0.9901$ . Row 1.0, col 0.00:  $P(Z \leq 1.00) = 0.8413$ .

$$P(1.00 \leq Z \leq 2.33) = 0.9901 - 0.8413 = \mathbf{0.1488}.$$

(d)  $P(|Z| \leq 1.96) = P(-1.96 \leq Z \leq 1.96) = 0.9750 - 0.0250 = \mathbf{0.9500}$ .

**3. Working backwards from the table body:**

- (a)  $P(Z \leq z^*) = 0.025$ . The table gives  $P(Z \leq -1.96) = 1 - 0.9750 = 0.0250$ . Therefore  $\mathbf{z^* = -1.96}$ .
- (b)  $P(Z > z^*) = 0.025 \Rightarrow P(Z \leq z^*) = 0.975$ . The table gives  $P(Z \leq 1.96) = 0.9750$ . Therefore  $\mathbf{z^* = +1.96}$ .

These two values,  $\pm 1.96$ , are the *critical values* for a 95% confidence level. They will appear in every two-tailed inference procedure in Chapters 5 and 6.

**4. Standardized exam scores  $Z \sim N(0, 1)$ .**

- (a)  $P(Z > 1.5) = 1 - P(Z \leq 1.5) = 1 - 0.9332 = \mathbf{0.0668}$ . About **6.68%** of students scored above  $z = 1.5$ .
- (b)  $P(-1 \leq Z \leq 1) = P(Z \leq 1) - P(Z \leq -1) = 0.8413 - (1 - 0.8413) = 0.8413 - 0.1587 = \mathbf{0.6826}$ . About **68.3%** of students scored between  $z = -1$  and  $z = 1$  (consistent with the Empirical Rule).
- (c) Top 15%:  $P(Z > z^*) = 0.15 \Rightarrow P(Z \leq z^*) = 0.85$ . Searching the table:  $P(Z \leq 1.03) = 0.8485$  and  $P(Z \leq 1.04) = 0.8508$ . The value 0.85 is closest to  $z = 1.04$  (difference 0.0008) versus  $z = 1.03$  (difference 0.0015). Therefore  $\mathbf{z^* \approx 1.04}$  is the minimum z-score needed to pass.

**5. Model response (writing prompt):**

The binomial probability histogram assigns a bar of width 1 to each integer outcome, so that bar height equals probability; as  $n$  grows large (with  $p$  fixed near  $1/2$ ), the bars multiply and narrow, and their outline traces an increasingly smooth, symmetric bell shape. In the limit, the bars become infinitesimally thin and their tops become a continuous curve: the normal density. The normal distribution is the natural limit of the binomial because, by the Central Limit Theorem, sums and counts of independent, identically distributed random variables — which is exactly what a binomial count is — always converge in distribution to a normal curve, regardless of the underlying probabilities, as long as  $n$  is large.